SDPM Project Proposal and Development Plan

1. Project Information

Project Title:Online Student Management System

Project Category Web Application

2. Team Information

Name	Student ID	Email	Primary Role
MD Ataus samad	2023000000059	2023000000059@seu.edu.bd	Project Manager & Full Stack Developer

3. Project Overview

The **Online Student Management System (OSMS)** is a web-based application designed to simplify and automate the management of student information in educational institutions. The system provides a centralized platform where administrators, teachers, and students can securely access and manage academic and administrative data. It includes features such as student registration, course enrollment, attendance tracking, grade management, and report generation.

The main objective of this project is to replace traditional paper-based record-keeping with an efficient digital solution that improves data accuracy, minimizes administrative workload, and enhances communication among users. By providing real-time access to student records, the system enables faster decision-making and better management of academic activities.

The project will be developed using the Agile Scrum methodology, allowing the development team to deliver the system in multiple iterations while incorporating stakeholder feedback throughout the process. The application will be built using modern web technologies with a secure authentication system, responsive user interface, and reliable database management to ensure usability, performance, and data security. Upon completion, the Online Student Management System is expected to improve operational efficiency, reduce manual errors, and provide a scalable platform that can support the future needs of educational institutions.

4. Problem Statement

- Many educational institutions continue to rely on manual or partially digital methods for managing student information, attendance, course enrollment, and academic records. These traditional approaches often result in data duplication, human errors, delayed access to information, and inefficient administrative processes. Managing large volumes of student data manually also makes it difficult to generate accurate reports and monitor academic performance effectively.

- The primary groups affected by this problem are students, teachers, and administrative staff. Students may experience delays in accessing their academic records and course information, while teachers spend significant time maintaining attendance and grade records. Administrative staff face challenges in organizing, updating, and retrieving student data efficiently, especially as the number of students increases.

- This problem is important because inefficient data management can reduce institutional productivity, increase operational costs, and negatively affect the quality of academic services. Accurate and timely information is essential for effective decision-making, communication, and student support.

- Although some institutions use existing software solutions, many are expensive, difficult to customize, or include unnecessary features that do not meet specific institutional requirements. Therefore, there is a need for a secure, user-friendly, and cost-effective Online Student Management System that simplifies academic administration while improving efficiency, data accuracy, and accessibility.

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5. Objectives

The main objectives of the Online Student Management System are:

- To develop a secure, user-friendly, and web-based system for managing student information and academic records.
- To automate student registration, course enrollment, attendance tracking, and grade management.
- To reduce manual paperwork, data duplication, and human errors in administrative processes.
- To improve the accuracy, consistency, and security of student data through centralized database management.
- To provide real-time access to academic information for students, teachers, and administrators.
- To enable quick generation of reports related to attendance, academic performance, and student records.
- To improve communication and coordination among students, teachers, and administrative staff.
- To enhance institutional efficiency by reducing processing time and simplifying routine academic management tasks.
- To develop a scalable and maintainable system that can be expanded with additional features in the future.

6. Proposed Solution

- The proposed solution is to develop a **web-based Online Student Management System (OSMS)** that automates the management of student information and academic activities through a centralized and secure platform. The system will replace manual record-keeping with digital processes, improving efficiency, accuracy, and accessibility.

- The overall workflow begins with secure user authentication. After logging in, administrators can register students, manage courses, assign teachers, and maintain student records. Teachers can record attendance, enter grades, and monitor student performance. Students can access their profiles, view course enrollments, attendance records, academic results, and other relevant information through their personal dashboards. All data will be stored in a centralized database, allowing users to retrieve and update information in real time while ensuring data consistency and security.

- The intended users of the system are **administrators, teachers, and students**. Each user role will have different access permissions to ensure that information is protected and only authorized users can perform specific tasks.

- The expected outcome is a reliable, user-friendly, and scalable system that reduces manual work, minimizes human errors, improves data accuracy, and enables faster decision-making. The solution will enhance academic administration, increase productivity, and provide a more efficient experience for both staff and students.

7. Scope

In Scope

The following features will be included in the project:

- User registration and secure login authentication
- Role-based access for administrators, teachers, and students
- Student profile management
- Course registration and management
- Attendance recording and tracking
- Grade entry and result management
- Student dashboard with personalized information
- Search functionality for student and course records
- Report generation for attendance and academic performance
- Database management with secure data storage
- Responsive web interface for desktop and mobile browsers

Out of Scope

The following features will not be implemented in this project due to time and resource constraints:

- Mobile application (Android/iOS)
- Online fee payment and payment gateway integration
- Video conferencing or virtual classroom features
- SMS and email notification services
- AI-based student performance prediction
- Integration with third-party learning management systems (LMS)
- Biometric attendance system
- Multi-language support
- Cloud deployment and distributed server infrastructure
- Advanced analytics and business intelligence dashboards

8. Functional Requirements (Features)

Feature	Description	Priority
User Registration & Login	Allows users to create accounts and securely log in to the system.	High
Role-Based Access Control	Provides different permissions for administrators, teachers, and students.	High
Student Profile Management	Enables administrators to create, update, and manage student records.	High
Course Management	Allows administrators to add, edit, and assign courses.	High
Course Enrollment	Enables students to enroll in available courses.	High
Attendance Management	Allows teachers to record and monitor student attendance.	High
Grade Management	Enables teachers to enter, update, and publish student grades.	High
Student Dashboard	Displays personal information, enrolled courses, attendance, and academic results.	Medium
Search Functionality	Allows users to quickly search for students, courses, and records.	Medium
Report Generation	Generates attendance reports, grade reports, and student summaries.	High
Password Management	Allows users to change or reset their passwords securely.	Medium
Responsive User Interface	Ensures the system works effectively on desktops, tablets, and mobile browsers.	Medium
Data Backup and Recovery	Protects system data through regular backups and recovery options.	Low
Activity Logs	Records important user actions for monitoring and auditing purposes.	Low

9. Technology Stack

Category	Technology
Frontend	HTML5, CSS3, JavaScript, Bootstrap 5
Backend	PHP
Database	Supabase
Programming Language(s)	PHP, JavaScript, HTML5, CSS3
Framework(s)	Bootstrap 5
Libraries/APIs (Must Use 1)	Chart.js (for visualizing attendance and academic performance reports)
Development Tools	Visual Studio Code, XAMPP, phpMyAdmin
Version Control	Git and GitHub
Types of Testing to be Used	Unit Testing, Integration Testing, System Testing, User Acceptance Testing (UAT), Functional Testing

10. System Architecture (Optional)

The Online Student Management System will follow a **three-tier architecture**, consisting of the Presentation Layer, Application Layer, and Database Layer.

- **Presentation Layer (Frontend):** Developed using HTML5, CSS3, JavaScript, and Bootstrap 5 to provide a responsive and user-friendly interface.
- **Application Layer (Backend):** Developed using PHP to process user requests, implement business logic, authenticate users, and communicate with the database through the Supabase API.
- **Database Layer: Supabase (PostgreSQL)** serves as the cloud-hosted database. It securely stores user accounts, student records, courses, attendance, grades, and other application data while providing scalable data management and API access.

System Workflow:

User → Web Browser → Frontend (HTML/CSS/JavaScript/Bootstrap) → PHP Backend → Supabase (PostgreSQL) → PHP Backend → Frontend → User

This architecture provides secure cloud-based data storage, reliable communication between the application and the database, and supports future scalability and maintenance.

11. Feasibility Analysis

Technical Feasibility

The Online Student Management System is technically feasible because it will be developed using technologies that are well understood and readily available, including HTML5, CSS3, JavaScript, Bootstrap, PHP, and Supabase (PostgreSQL). These technologies are widely used for web application development and are supported by extensive documentation and community resources. The project can be developed using the current knowledge and skills of the development team while also providing opportunities to learn and apply modern cloud database technologies.

Time Feasibility

The project is feasible to complete within the semester by following an Agile Scrum development approach. The work will be divided into manageable sprints covering requirement analysis, system design, implementation, testing, and deployment. Regular progress reviews and milestone tracking will help ensure that the project is completed within the available academic timeline.

Resource Feasibility

The project requires a standard personal computer with an internet connection for development and testing. The software resources include Visual Studio Code, Git, GitHub, XAMPP, and the Supabase cloud platform for database management. The system will use the Supabase REST API to communicate with the database and Chart.js to visualize reports. No paid software, specialized hardware, or proprietary datasets are required, making the project cost-effective and practical to implement.

12. Work Distribution

Clearly define the responsibilities of each team member.

Team Member	Responsibilities
MD Ataus Samad	Project planning and management; requirements gathering and analysis; system design; UI/UX design; frontend development (HTML, CSS, JavaScript, Bootstrap); backend development (PHP); database design and integration using Supabase (PostgreSQL); API integration; testing (Unit, Integration, System, and User Acceptance Testing); bug fixing; version control using Git and GitHub; project documentation; deployment and final presentation.

13. Development Timeline

Week	Planned Activities
Week 2	Project planning, requirement gathering, problem analysis, and proposal preparation.

Week 3	System analysis, software requirements specification (SRS), database design, and UI wireframe creation.
Week 4	Set up the development environment, configure Supabase, and implement user authentication (registration and login).
Week 5	Develop the student management module, including student registration, profile management, and CRUD operations.
Week 6	Implement course management, course enrollment, and role-based access control for administrators, teachers, and students.
Week 7	Develop attendance management, grade management, and the student dashboard.
Week 8	Implement search functionality, report generation, and integrate the frontend with the Supabase database and APIs.
Week 9	Conduct unit testing and integration testing, fix identified bugs, and optimize system performance.
Week 10	Perform system testing and User Acceptance Testing (UAT), improve the user interface, and finalize system features.
Week 11	Prepare project documentation, user manual, and deployment guide; deploy the application for demonstration.
Week 12	Final project review, presentation preparation, submit the completed project and documentation, and address supervisor feedback if required.

14. Risks and Challenges

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Risk	Proposed Mitigation
Requirement changes during development	Follow Agile methodology, review requirements regularly, and update the project plan when necessary.
Limited development time	Create a realistic schedule, prioritize high-priority features, and monitor weekly progress.
Technical issues with Supabase integration	Study the official documentation, test API connections early, and seek guidance when needed.
Bugs and software defects	Perform regular unit, integration, and system testing throughout development.
Data loss or database errors	Enable database backups and use Git/GitHub for source code version control.
Internet connectivity issues	Keep local backups of project files and continue offline development whenever possible.
Security vulnerabilities	Implement secure authentication, validate user inputs, and apply proper access control.
Lack of user feedback	Conduct demonstrations with the supervisor or classmates and incorporate feedback before final submission.

15. Expected Deliverables

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By the end of the semester, the project team expects to deliver the following:

- A fully functional **Online Student Management System** web application.
- Complete source code with proper documentation and comments.
- A **Supabase (PostgreSQL)** database containing the required tables, relationships, and sample data.

- User authentication and role-based access control for administrators, teachers, and students.
- System documentation, including software architecture, database design, and API integration details.
- User documentation (User Manual) explaining how to use the system.
- Test reports covering unit, integration, system, and user acceptance testing.
- Project documentation, including the Software Requirements Specification (SRS), design documents, and implementation report.
- GitHub repository containing the complete project with version history.
- Final project report submitted according to course requirements.
- Project presentation and live demonstration of the completed system.

16. References (Optional)

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8. GitHub. *GitHub Documentation*. https://docs.github.com/
9. Chart.js. *Chart.js Documentation*. https://www.chartjs.org/docs/latest/
10. Agile Alliance. *Agile Methodology Resources*. https://agilealliance.org/

Declaration

We certify that this proposal represents our own planned work. We understand that significant deviations from this proposal during development should be discussed with the course instructor.

Team Members' Names:

1. **MD ATAUS SAMAD**